Avoiding Cavitation: Problem 5–63

Worked solution

Problem. Water flows from an open reservoir through a horizontal pipe. The free surface (3) is 5 m above sections (1) and (2), and the outlet diameter is $D_2 = 15\,\text{cm}$. Find the minimum diameter at section (1) that avoids cavitation there.

Assumptions. The flow is steady and incompressible, losses are negligible, and the reservoir is large enough that $V_3 \approx 0$. Both the free surface and outlet are at atmospheric pressure. Use $\rho = 1000\,\text{kg/m}^3$, $g = 9.81\,\text{m/s}^2$, and $p_{\text{atm}} = 101.3\,\text{kPa}$. All pressures are absolute. The temperature is not given; assume 20°C, for which $p_v \approx 2.34\,\text{kPa}$.[1]

1 Outlet velocity

Take $z_1 = z_2 = 0$ and $z_3 = 5\,\text{m}$. Bernoulli's equation between the free surface (3) and outlet (2) gives

$$\frac{p_3}{\rho g} + \frac{V_3^2}{2g} + z_3 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2.$$

Since $p_3 = p_2 = p_{\text{atm}}$ and $V_3 \approx 0$, this reduces to

$$5 = \frac{V_2^2}{2g}, \qquad V_2 = \sqrt{2g(5\,\text{m})} \approx 9.90\,\text{m/s}.$$

2 Cavitation limit at section (1)

Cavitation begins when the local absolute pressure reaches the vapor pressure. Set $p_1 = p_v$ to find the limiting velocity. Bernoulli between (1) and (2), which are at the same elevation, yields

$$\frac{p_v}{\rho g} + \frac{V_{1,\max}^2}{2g} = \frac{p_{\text{atm}}}{\rho g} + \frac{V_2^2}{2g}.$$

Hence

$$V_{1,\max} = \sqrt{V_2^2 + \frac{2(p_{\text{atm}} - p_v)}{\rho}} = \sqrt{98.1 + \frac{2(101300 - 2340)}{1000}}\,\text{m/s} \approx 17.2\,\text{m/s}.$$

3 Minimum diameter

For circular sections, continuity gives

$$A_1 V_1 = A_2 V_2, \qquad \frac{\pi D_1^2}{4} V_1 = \frac{\pi D_2^2}{4} V_2.$$

The smallest allowable diameter corresponds to $V_1 = V_{1,\max}$. Thus

$$D_{1,\min} = D_2 \sqrt{\frac{V_2}{V_{1,\max}}} \approx 0.15 \sqrt{\frac{9.90}{17.2}}\,\text{m} \approx \boxed{11.4\,\text{cm}}.$$

If vapor pressure is neglected ($p_v \approx 0$), the same calculation gives $D_{1,\min} \approx 11.3\,\text{cm}$. A smaller diameter raises the velocity and lowers the pressure at (1). Equality marks the onset of cavitation; a diameter above the limiting value keeps $p_1 > p_v$ under the stated assumptions.

[1] NIST, *Vapor pressure formulation for water in range 0 to 100 degrees C: A revision*, Table 3.